

AI-Driven Customer Service Chatbot & Contact Center Automation

Client: Major Regional Telecommunications Operator — Confidential

Supplier: NileTech Solutions

Document Classification: Historical Technical Proposal — Won

Proposal Reference: NTS-TEL-AICC-091/2025

Delivery Model: Hybrid Agile / MLOps / Controlled AI SDLC

Deployment Model: Azure Private / Enterprise Cloud

Primary Channels: Web, Mobile App, WhatsApp, Contact Center

Programme Duration: 36 Weeks

Target Languages: Arabic, Egyptian Arabic, English

AI Architecture: Enterprise RAG + LLM Gateway + Tool Calling

1. Executive Summary & Project Scope

1.1 Executive Summary

NileTech Solutions was awarded the implementation of an **AI-driven customer service platform** designed to automate first-line customer interactions across the telecommunications operator's digital channels.

The existing customer-service model relied heavily on conventional FAQ systems, static decision trees and contact-center agents. This resulted in:

- High volumes of repetitive Level-1 customer enquiries.
- Long handling times for common service requests.
- Fragmented knowledge across billing, network, product and policy systems.
- Inconsistent responses between digital channels and contact-center agents.
- Limited support for **Egyptian Arabic / Arabic-English code switching**.
- High escalation rates for questions requiring information from multiple backend systems.

The proposed solution combined **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **enterprise APIs** and **controlled tool execution**.

The chatbot was explicitly designed **not to operate as an unrestricted generative assistant**.

For regulated or customer-impacting workflows, responses were required to be grounded in approved enterprise knowledge and/or retrieved customer-specific system data.

1.2 Business Objectives

The programme objectives were:

- 1. Automate high-volume Level-1 customer-service enquiries.
- 2. Provide consistent answers across digital channels.
- 3. Support **Arabic, Egyptian Arabic and English**.
- 4. Ground informational responses using enterprise-approved knowledge.
- 5. Allow controlled access to customer-specific information.
- 6. Reduce unnecessary human-agent escalation.
- 7. Provide complete conversational auditability.
- 8. Detect low-confidence responses before they reach customers.
- 9. Integrate with CRM, billing, provisioning and ticketing systems.
- 10. Establish an enterprise **AI governance and MLOps framework**.

1.3 Functional Scope

Customer-Facing Capabilities

Capability	Description	Data Source
Account Information	Account status and customer profile information	CRM
Billing Queries	Current balance, invoices and payment status	Billing
Package Information	Package details, limits and eligibility	Product Catalogue
Usage	Internet/data/voice usage information	Charging System
Service Requests	Selected service activation/deactivation	Provisioning APIs
Troubleshooting	Guided technical troubleshooting	Knowledge Base
Outage Information	Service/network incident information	Network OSS
Complaints	Complaint creation and status	CRM / Ticketing
FAQ	Policy and product questions	RAG Knowledge Base
Human Escalation	Transfer to contact center	Contact Center Platform

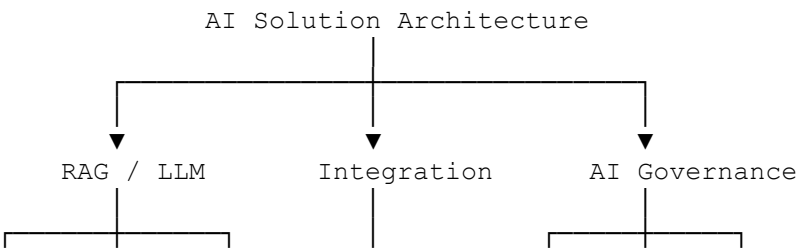
2. Team Composition & Resourcing

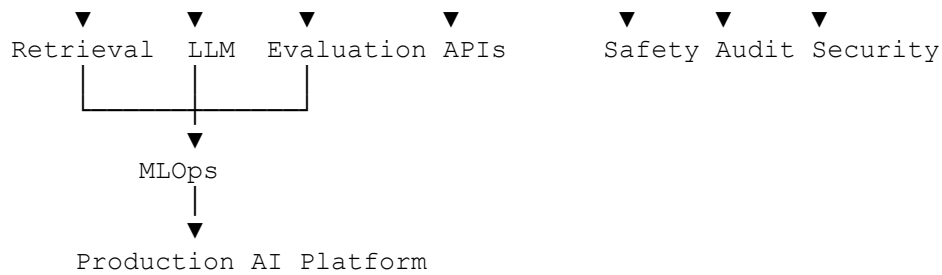
NileTech assembled a multidisciplinary AI delivery team covering **LLM engineering, RAG, data engineering, cybersecurity, telecom integration, MLOps and conversational UX**.

2.1 Core Delivery Team

Role	Years of Exp	Certifications	Allocation %
Programme Director	18	PMP, PRINCE2 Practitioner	20%
AI Solution Architect	14	TOGAF, Azure AI Engineer Associate	100%
Telecom Domain Architect	16	TM Forum Certified	60%
AI Product Manager	11	PMI-ACP, PSPO	100%
Lead GenAI Engineer	10	Azure AI Engineer, NVIDIA GenAI	100%
Senior LLM Engineers ×3	5–9	Azure AI / Hugging Face	100%
RAG Engineer ×2	5–8	Azure AI Engineer	100%
ML Engineer ×2	6–9	TensorFlow / Azure ML	80%
Data Engineering Lead	12	Azure Data Engineer	80%
Data Engineers ×2	5–8	DP-203	100%
MLOps Engineer ×2	6–9	Azure ML, Docker, Kubernetes	100%
Integration Lead	13	Azure Integration Services	80%
Integration Engineers ×3	5–9	Azure API Management	100%
Conversation Designer ×2	6–9	Conversation Design Certification	100%
AI Security Engineer	10	CISSP, CCSP	60%
Responsible AI Specialist	9	Microsoft Responsible AI	50%
QA / AI Evaluation Lead	10	ISTQB Advanced	100%
AI Evaluation Engineers ×2	4–7	ML Evaluation / ISTQB	100%
Performance Engineer	8	ISTQB Performance Testing	40%
DevSecOps Lead	11	AZ-400, CKA	80%
Business Analysts ×2	7–11	CBAP	100%

2.2 AI Workstream Structure

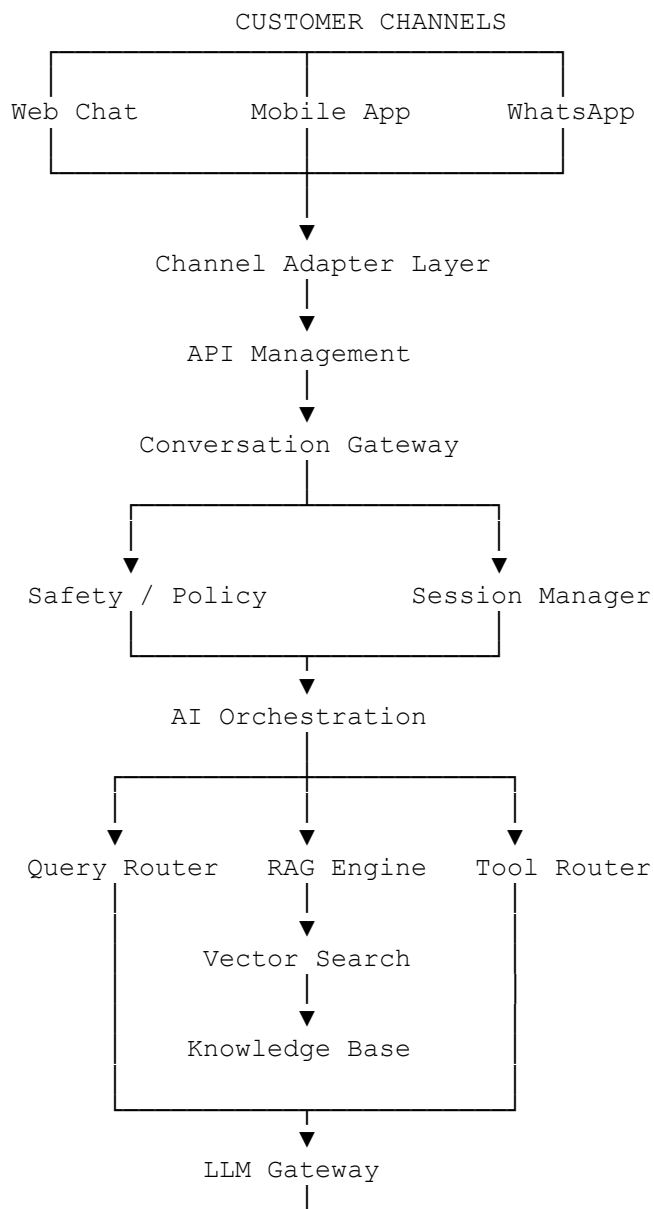


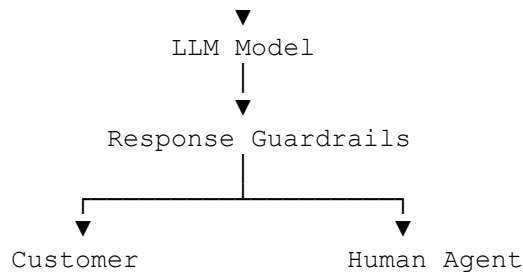


3. Technical Architecture & Tech Stack

3.1 Target Architecture

The solution was designed as a **controlled RAG-based conversational AI platform**.

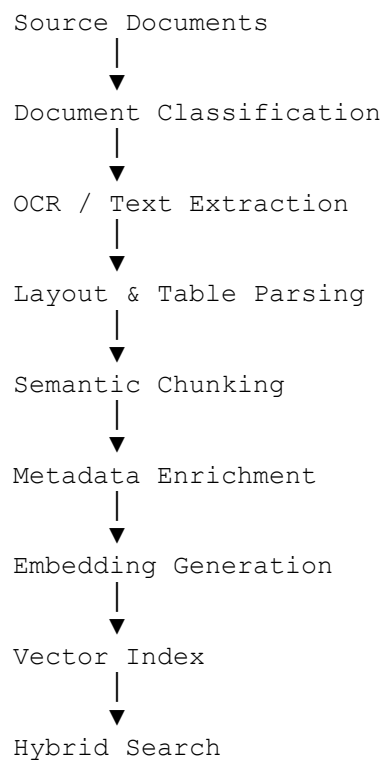




3.2 Retrieval-Augmented Generation Pipeline

The RAG architecture was designed to separate **knowledge retrieval** from **generation**.

Ingestion Pipeline



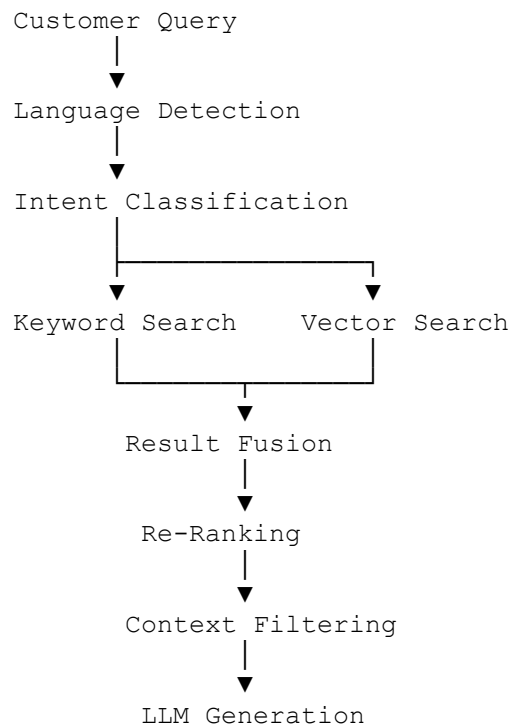
Documents were enriched with metadata including:

- Document ID.
- Document version.
- Product.
- Service category.
- Language.
- Effective date.
- Expiration date.

- Customer segment.
 - Regulatory classification.
 - Source system.
 - Access classification.
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3.3 Retrieval Strategy

The platform used **hybrid retrieval** rather than relying exclusively on vector similarity.



Retrieval Controls

- Top-K retrieval.
 - Metadata filtering.
 - Semantic similarity threshold.
 - Cross-encoder / reranking.
 - Duplicate-context elimination.
 - Document-version filtering.
 - Source authority weighting.
 - Effective-date validation.
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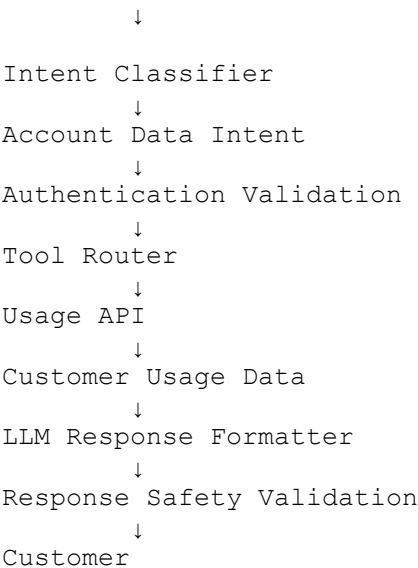
3.4 Enterprise Tool Calling

Customer-specific operations were not answered solely from the vector database.

The AI orchestration layer could invoke approved enterprise APIs.

Example:

Customer:
"What is my current internet usage?"



Available tools included:

Tool	Function	Authorization
get_customer_profile	Retrieve customer information	Authenticated
get_current_balance	Retrieve account balance	Authenticated
get_usage	Retrieve usage statistics	Authenticated
get_active_package	Retrieve active package	Authenticated
create_service_request	Create approved request	Authenticated + confirmation
create_complaint	Create CRM complaint	Authenticated + confirmation
check_outage	Check network incident	Public / Location-dependent
create_human_escalation	Transfer conversation	Session-level

3.5 Technology Stack

Layer	Technology	Purpose
Channel Layer	Web / Mobile / WhatsApp	Customer interaction

Layer	Technology	Purpose
API Gateway	Azure API Management	API governance
Backend	Python / FastAPI	AI orchestration APIs
AI Orchestration	LangGraph / custom orchestration	Stateful agent workflow
LLM Gateway	Enterprise LLM Gateway	Model abstraction and governance
Foundation Model	Enterprise-approved LLM	Natural-language generation
Embeddings	Azure/Open-source embedding model	Semantic retrieval
Vector Store	Azure AI Search	Vector + hybrid retrieval
Reranking	Semantic Ranker	Retrieval refinement
Document Processing	Azure AI Document Intelligence	OCR/layout extraction
Storage	Azure Blob Storage	Source documents
Metadata	PostgreSQL	Configuration and audit metadata
Cache	Redis	Session / response caching
Messaging	Azure Service Bus	Asynchronous workloads
ML Platform	Azure Machine Learning	Evaluation/MLOps
Containers	Docker / AKS	Deployment
Secrets	Azure Key Vault	Secret management
CI/CD	Azure DevOps	Deployment pipelines
Monitoring	Azure Monitor / App Insights	Observability
SIEM	Microsoft Sentinel	Security monitoring
Analytics	Power BI	AI/business analytics

3.6 AI Safety & Guardrails

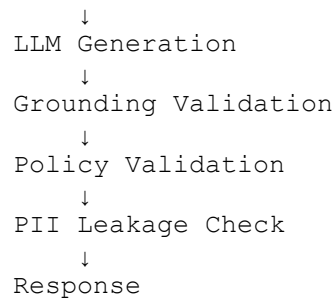
Because the platform interacted directly with telecom customers, unrestricted generation was prohibited.

The response pipeline included:

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User Input
  ↓
Prompt Injection Detection
  ↓
PII / Sensitive Data Detection
  ↓
Intent Classification
  ↓
Authorization
  ↓
Retrieval / Tool Execution

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Mandatory AI Controls

- **No unsupported factual claims** for regulated/product-policy responses.
- No disclosure of internal system prompts.
- No disclosure of internal API credentials or implementation details.
- Customer-specific information only after authentication.
- Tool execution restricted through an allowlist.
- Transactional actions required explicit confirmation.
- Low-confidence responses required escalation or controlled fallback.
- All AI interactions required traceability.
- Prompt and response logging subject to PII masking policies.

4. Implementation Timeline

The implementation was structured as a **36-week programme** with AI engineering, enterprise integration and evaluation workstreams operating in parallel.

Phase	Deliverables	Duration in Weeks	Dependencies
1. Mobilisation	Governance, AI programme charter, environments, stakeholder map	2	Contract award
2. Discovery & Use-Case Definition	Intent catalogue, channel assessment, automation candidates	4	Business SMEs
3. Knowledge Assessment	Document inventory, source authority matrix, content quality assessment	4	Knowledge owners
4. AI Architecture	HLD, RAG architecture, model strategy, guardrail architecture	4	Discovery
5. Data & Document Pipeline	OCR, parsing, chunking, metadata and indexing pipelines	6	Knowledge assessment
6. RAG MVP	Retrieval, vector search, hybrid search, reranking	6	Document pipeline

Phase	Deliverables	Duration in Weeks	Dependencies
7. Conversational AI	Intent routing, conversation state, prompt orchestration	6	RAG MVP
8. Enterprise Integrations	CRM, billing, usage, package and ticketing APIs	8	API availability
9. AI Guardrails	Safety filters, grounding checks, PII protection	5	Conversational AI
10. MLOps Platform	Evaluation pipelines, model registry, CI/CD, observability	5	AI architecture
11. AI Evaluation	Golden datasets, retrieval metrics, response evaluation	4	RAG MVP
12. SIT	End-to-end channel/API/AI testing	4	Integration completion
13. Security Testing	Prompt injection, API security, penetration testing	3	Feature freeze
14. Performance Testing	Concurrency, latency, throughput and load testing	3	Stable release
15. UAT / Business Pilot	Controlled customer cohort, business acceptance	4	SIT + Security
16. Production Rollout	Phased deployment and hypercare	2	UAT approval
17. Handover	Training, runbooks, documentation, operational acceptance	2	Production stabilization

4.1 AI Evaluation Framework

The programme did not use **LLM output quality alone** as the acceptance criterion.

The evaluation framework measured separate dimensions.

Metric	Definition	Target
Recall@K	Relevant source retrieved within Top-K	≥ 90%
Precision@K	Retrieved context relevance	≥ 85%
MRR	Ranking quality of relevant source	≥ 0.85
Context Precision	Relevant context proportion	≥ 85%
Context Recall	Required evidence coverage	≥ 90%
Faithfulness	Answer supported by retrieved evidence	≥ 95%
Answer Relevance	Response relevance to user query	≥ 90%
Unsupported Claim Rate	Answers containing unsupported assertions	≤ 2%
Escalation Accuracy	Correct identification of human-required cases	≥ 95%

Important acceptance condition: aggregate model quality could not compensate for failure of critical safety or grounding controls.

4.2 Golden Dataset Structure

The evaluation dataset was divided into controlled categories:

Category	Test Cases	Examples
Product FAQ	1,200	Packages, pricing, eligibility
Billing	800	Invoices, balances, payment issues
Technical Support	1,000	Connectivity and troubleshooting
Account	500	SIM/account/profile
Complaints	400	Service complaints
Arabic	1,000	Modern Standard Arabic
Egyptian Arabic	800	Colloquial/code-switched queries
English	600	Standard English
Adversarial	500	Prompt injection / manipulation
Out-of-Domain	400	Unsupported requests

Total baseline evaluation corpus: 8,200 test cases

5. SLA & Support Matrix

The production service was governed by a severity-based SLA covering the **AI platform, integrations, retrieval infrastructure and customer-facing channels**.

Severity	Definition	Initial Response	Technical Engagement	Target Workaround	Target Restoration
Severity 1 — Critical	Complete chatbot outage, widespread incorrect transactions, critical security/AI safety incident	15 min	≤30 min	1 hour	4 hours
Severity 2 — High	Major channel degradation, significant integration failure or elevated incorrect-response rate	30 min	≤1 hour	4 hours	8 hours

Severity	Definition	Initial Response	Technical Engagement	Target Workaround	Target Restoration
Severity 3 — Medium	Limited intent failure, isolated integration issue or degraded retrieval quality	2 hours	≤4 hours	1 Business Day	3 Business Days
	Minor conversational defect, wording issue, low-impact knowledge gap	4 Business Hours	≤1 Business Day	N/A	5 Business Days / Planned Release

5.1 AI-Specific Incident Classification

AI incidents were additionally classified according to operational risk.

AI Incident	Example	Severity
Systemic Hallucination	Model provides unsupported billing policy to large customer population	S1
Customer Data Leakage	Customer A receives information belonging to Customer B	S1
Unauthorized Tool Execution	AI invokes restricted customer operation	S1
Knowledge Version Failure	Expired tariff/policy retrieved	S2
Retrieval Degradation	Significant reduction in retrieval quality	S2
Language Quality Issue	Arabic/Egyptian Arabic misunderstanding	S3
Minor Response Formatting	Incorrect formatting without factual impact	S4

5.2 AI Operations Monitoring

The production platform monitored both **traditional application telemetry** and **AI-specific operational metrics**.

Platform Metrics

- CPU / memory utilisation.
- API latency.
- Error rates.
- Request throughput.

- Queue depth.
- Database latency.
- Vector search latency.
- Token consumption.

AI Metrics

- Retrieval hit rate.
 - Average retrieval score.
 - Reranker score distribution.
 - Grounding failures.
 - Hallucination/unsupported-claim indicators.
 - Tool-call failure rate.
 - Tool-call rejection rate.
 - Human escalation rate.
 - Prompt-injection detection rate.
 - Response latency.
 - Model error rate.
 - Token usage per conversation.
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6. Compliance & AI Governance

The proposed AI platform implemented a formal **AI Governance Framework**.

6.1 Model Governance

Every production model required:

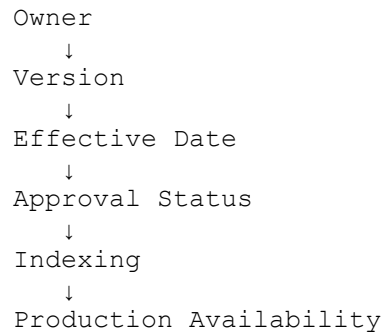
- Model/version identification.
 - Approved use-case classification.
 - Evaluation dataset.
 - Benchmark results.
 - Security assessment.
 - Bias/fairness assessment where applicable.
 - Prompt/version control.
 - Rollback model/version.
 - Production approval.
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6.2 Knowledge Governance

Every enterprise knowledge source required:

Document

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Expired or superseded documents were automatically excluded from production retrieval where metadata allowed deterministic filtering.

6.3 Human Escalation Policy

The chatbot was required to transfer conversations to human agents when:

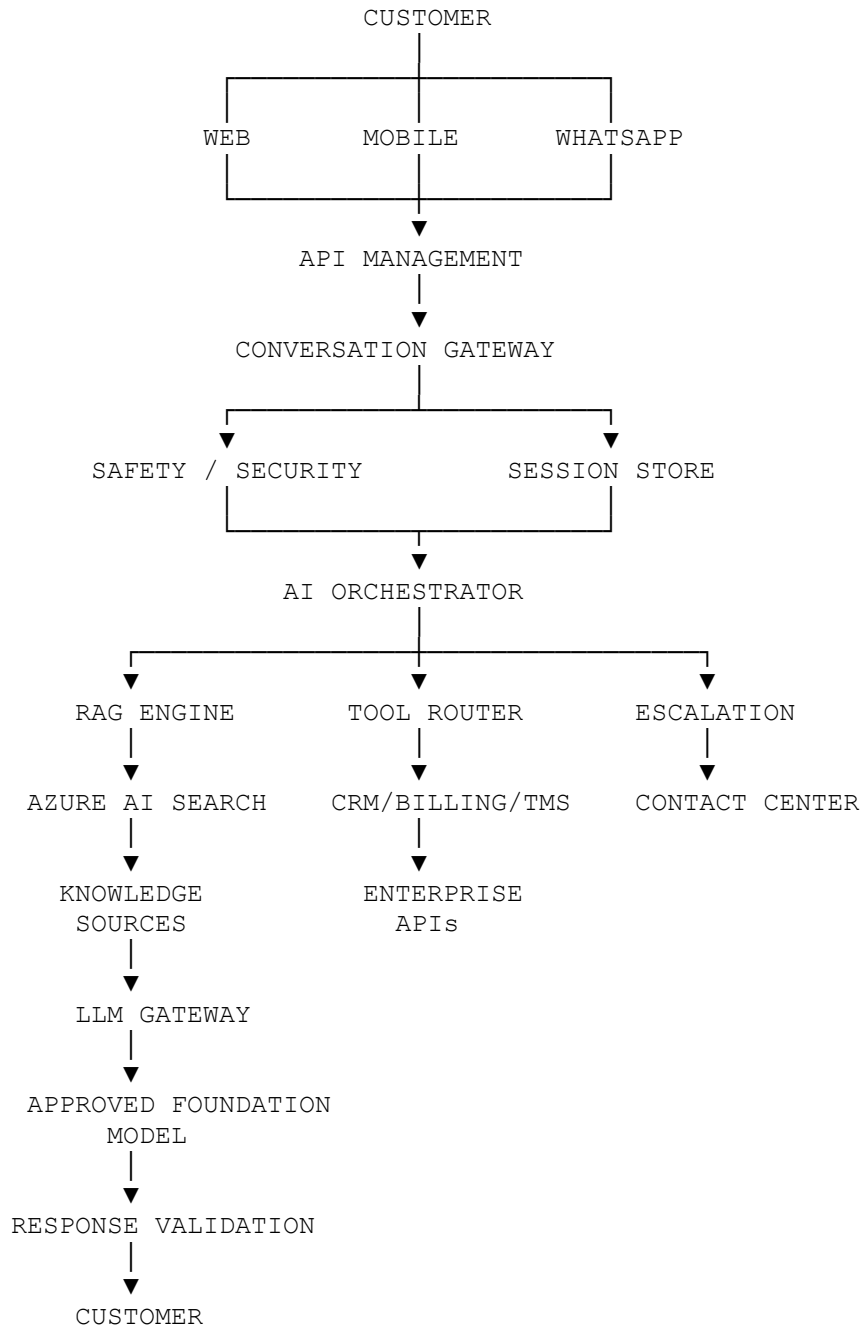
- Customer explicitly requested an agent.
- Confidence fell below the configured threshold.
- Required information was unavailable.
- A regulated or restricted action could not be safely automated.
- The customer demonstrated repeated misunderstanding.
- The system detected a potential security or account-takeover scenario.
- The request required discretionary human judgment.

Escalation Payload

The contact-center agent received:

Field	Description
Customer ID	Authenticated customer identifier
Conversation ID	Unique session reference
Intent	Classified customer intent
Conversation Summary	AI-generated summary
Retrieved Sources	Knowledge references used
Tool Calls	Enterprise APIs invoked
Failed Attempts	Relevant failed interactions
Escalation Reason	Deterministic escalation category
Language	Arabic / Egyptian Arabic / English
Sentiment Indicator	Operational routing signal

7. Production Architecture — Final State



8. Final Acceptance Criteria

Production acceptance required all of the following:

Gate	Acceptance Requirement
Architecture Gate	Approved AI, integration and security architecture
Knowledge Gate	Approved production knowledge corpus
Retrieval Gate	Retrieval benchmark achieved
Grounding Gate	Critical-domain faithfulness threshold achieved
Security Gate	No unresolved Critical security findings
AI Safety Gate	Prompt-injection and unauthorized-tool tests passed
Integration Gate	CRM/billing/provisioning integrations accepted
Performance Gate	Agreed concurrency and latency targets achieved
UAT Gate	Business stakeholders approve target journeys
Operational Gate	Monitoring, alerting and escalation operational
Human Handoff Gate	Contact-center escalation workflow validated
MLOps Gate	Model/version rollback capability validated

Historical Programme Outcome

NileTech Solutions delivered the telecom operator's transition from **rule-based digital customer service toward a governed enterprise conversational-AI platform**, combining RAG, hybrid retrieval, enterprise API tool execution, multilingual conversation handling, automated evaluation and human-agent escalation.

The final architecture deliberately separated:

Knowledge Retrieval $\neq$ Customer Data Retrieval $\neq$ Transaction Execution $\neq$ Language Generation

This separation was established as a core architectural control to prevent the LLM from becoming an uncontrolled system-of-record or transaction authority.

Historical Proposal Status: Awarded / Production Deployment Completed / AI Operations Transitioned to Managed Support.